

NBA Fantasy Project

Overview

This is to build a player rankings and projections tool to help with my nba fantasy standard 9 cat, daily changes, weekly head to head league. The categories are Points (PTS), Rebounds (REB), Assists (AST), Steals (STL), Blocks (BLK), Three-pointers made (3PM), FG%, FT%, Turnovers (TO) (turnovers negative). Both percentage categories should be weighted by volume. When referring to the categories in chat use the full name, in anything we output it should be the abbreviation in brackets.

This is a hobby project and I want to keep it free as it is just for fun. I have never done any coding so while I want to use the best tools being a beginner needs to be taken into consideration. It is a brand new project, no work has been done on it at all.

I want to use the `nba_api` for the dataset. I will start with a basic z score for the fantasy ranking but plan to expand into more advanced/ specific ranking methods as I build it out to overcome the limitations of z score.

Scope

There are 3 parts to this project

1. A player ranking system that can be used for the draft or to see who the best performers over a period have been. Options for the period should be a drop down showing 1 week, 2 weeks, 4 weeks, full season. This should be sortable and be able to 'punt' categories and have that be taken into consideration for the ranking.
2. A projection tool that will allow me to tweak input fields such as usage, minutes, or stat rate (ie rebound rate) to show the players expected stats and therefore fantasy ranking based on those changes.
3. Being linked to my league so that it knows what players are rostered to each team. This will allow me to compare my team with my opponent for the week as well as show what players are free agents and therefore are the best available to be picked up. This can start off as a manual input but being able to pull this information would be a good future iteration.

Tools like basketball monster and hashtag basketball are good examples to work off.

I should also be able to filter when viewing players. Filters should include, position and the team they play for.

Another good future iteration will be to link to the nba schedule so that I can see who is playing today and take number of games played in a week into consideration.

Phasing

I expect to start with an offline document which has pulled real data from the nba and has some sort of fantasy scoring system. I will then build on this to tweak the fantasy scoring/ ranking method until the rankings start to reflect what I would expect.

An ability to manually show a player as rostered should come next. This can start as a simple rostered yes/ no and then grow into showing what specific team the player is on.

Once I am happy that I am able to pull data and the fantasy ranking method is good enough I will then start with the projection tool.

When the baseline of the two main aspects of this tool are set then I will look into automatically pulling the nba data so that I am working on up to date stats.

Audience

Just me, I have no plans to publish any of this work to anyone else.

Tech approach

Python, using `nba_api` for data. Interface built with Streamlit (simple local web app, no separate frontend framework). Data stored as local CSV files for now.

Working style

I'm new to coding and to working with AI on projects like this. Please explain your reasoning as we go, not just hand over code. Flag it if a request seems to contradict earlier decisions in this project rather than just going along with it.